

LocalLoop

Community Exchange Platform

Software Requirements Specification (SRS)

System Architecture & Technology Framework

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GitHub: github.com/RajputSivam/LocalLoop-Community-Exchange-Platform

1. Introduction

1.1 Project Overview

LocalLoop is a peer-to-peer community exchange platform that allows users to borrow, lend, and exchange items and skills securely within their local community. The platform is designed with FAANG-level system thinking, incorporating features such as multi-user exchange chains, AI-assisted negotiation, secure deposits, safe exchange zones, real-time tracking, and comprehensive analytics.

1.2 Purpose of this Document

This Software Requirements Specification (SRS) defines the functional and non-functional requirements, system architecture, and technology stack for LocalLoop. It serves as the primary reference document for development, testing, and project evaluation.

1.3 Project Scope

The platform covers the full lifecycle of a community exchange transaction:

- Item and skill listing by owners
- Borrow request initiation and agreement handling
- Security deposit locking via an in-app wallet
- Multi-user exchange chain detection and resolution
- Safe exchange zone recommendation using geolocation
- Return tracking with proof-of-condition (image/video)
- Late return and damage handling with automated penalties
- AI-assisted dispute resolution with admin oversight
- Trust scoring and rating system
- Real-time notifications via WebSocket / Firebase
- User and admin analytics dashboards

1.4 Stakeholders

Stakeholder	Role
End Users (Borrowers/Lenders)	List, borrow, exchange items and skills
Platform Admin	Manage disputes, monitor fraud, access analytics
System AI Module	Negotiate trades, compare conditions, suggest verdicts
Wallet Service	Lock/release security deposits, track transactions

2. Functional Requirements

The following table describes all functional requirements for LocalLoop:

ID	Feature	Description
FR-01	User Registration & Profile Management	Users can register, create profiles, set trust levels, and manage their exchange history.
FR-02	Item & Skill Listing	Owners can list items/skills with description, condition, images, borrow duration, and damage rules.
FR-03	Borrow Request with Agreement	Borrowers can request items, view deposit requirements and terms, and accept digital agreements.
FR-04	Security Deposit / Wallet Integration	System locks borrower's deposit dynamically based on item value and trust score.
FR-05	Multi-User Exchange Chain Detection	System detects circular or chain exchange opportunities (A has B's item, B has C's, C has A's).
FR-06	Safe Exchange Zone Recommendation	Google Maps integration suggests verified public exchange locations near both parties.
FR-07	AI Negotiation Assistant	LLM-powered module suggests trade alternatives, counter-offers, and fair deal assessments.
FR-08	Return Tracking & Proof of Condition	Borrowers submit image/video proof; system compares against original listing condition.
FR-09	Late / Damage Handling & Penalties	Overdue items trigger reminders and penalties; damaged items trigger proportional deposit deductions.
FR-10	Dispute Resolution System	Users submit evidence; AI suggests verdict; admin makes final decision with full audit trail.
FR-11	Ratings & Trust System	Both parties are rated post-exchange; penalties and praises affect trust score.
FR-12	Real-Time Notifications	WebSocket/Firebase sends real-time alerts for requests, approvals, overdue items, and disputes.
FR-13	User & Admin Dashboard	Users see active borrowings and history; admins see exchange stats, dispute metrics, and fraud flags.
FR-14	Multilingual AI Chatbot	Always-on LLM-powered chat assistant supporting 17 languages, live data integration, intent detection, and human handoff for complex queries.
FR-15	Real-Time Language Translation	Instantly translate messages between users with AI auto-detection of source language; both original and translated text stored.
FR-16	Voice Input & Output	Speech-to-text input via Whisper API; text-to-speech output via Google TTS / ElevenLabs in the user's preferred language.
FR-17	Personalized Recommendations	AI-driven suggestions for items, posts, local events, and services based on user activity vectors and collaborative filtering.
FR-18	Chat Summarization	LLM-generated summaries of long conversations or articles in the user's language, surfaced as 3–5 bullet points on demand.
FR-19	Context-Aware Responses	AI retains user preferences and interaction history (consent-gated) to personalise all chatbot and assistant responses.
FR-20	Emoji, GIF & Sticker Support	Unicode emoji picker, GIPHY-powered GIF search, and custom LocalLoop sticker packs within the chat interface.

ID	Feature	Description
FR-21	Image & Media Recognition	OpenAI Vision API identifies objects, landmarks, item conditions, and extracts/translates text from user-uploaded images.
FR-22	Quick Reply Suggestions	AI generates contextual reply chips and action shortcuts beneath incoming messages for faster, one-tap responses.
FR-23	Community Threads & Polls	Topic-based discussion threads and polls with AI-generated summaries of responses and majority-opinion highlights.
FR-24	Sentiment & Mood Detection	Real-time sentiment scoring of messages; empathy mode activates for distressed users with adapted, supportive replies.
FR-25	Gamification & Rewards	XP points, badges, weekly challenges, and a real-time leaderboard to drive engagement and reward positive exchange behaviour.
FR-26	Accessibility Options	Dark/light mode, adjustable font sizes, high-contrast theme, keyboard navigation, ARIA compliance, and voice command control.
FR-27	Safety & Moderation	AI moderation pipeline scans messages and images before delivery; blocked content queued for admin review; report/block controls.
FR-28	Offline Chat Mode	Service Worker caches recent conversations and FAQ responses; offline-composed messages synced on reconnection.
FR-29	Multi-Platform Sync	Firebase Firestore syncs conversations, preferences, and state across web, Android, and iOS within 200ms.
FR-30	Smart Notifications	AI personalises notification frequency and urgency per user, suppressing low-priority alerts and surfacing critical ones instantly.
FR-31	AI-Powered FAQs & Help	Semantic search over an embedded knowledge base provides instant, accurate platform answers without a support ticket.
FR-32	Topic-Based Chatrooms	Interest- or location-based rooms with AI-suggested active rooms and AI-generated summaries for new joiners.
FR-33	Content Moderation with Auto-Tagging	AI auto-tags all posts and images for discovery; proactively flags policy violations before content is publicly visible.
FR-34	Collaborative Tools	Shared notes, to-do lists, and lightweight documents within community groups with real-time co-editing via Yjs CRDT.
FR-35	JWT / OAuth 2.0 Authentication	Stateless authentication via RS256-signed JWT tokens with 15-minute expiry and rotating refresh tokens; OAuth 2.0 social login supported.
FR-36	Multi-Factor Authentication	Email OTP + SMS as second factor; TOTP as optional third factor; mandatory for all admin accounts.
FR-37	Role-Based Access Control	Three-tier RBAC (User, Moderator, Admin) enforced at API middleware level with distinct permission sets per role.
FR-38	Data Encryption at Rest & Transit	AES-256-GCM for PII fields at rest; TLS 1.3 for all communication; mTLS for internal service calls.
FR-39	Password Security	Argon2id hashing (OWASP params); compromised password check via HaveIBeenPwned API on registration and change.
FR-40	Rate Limiting & Input Validation	Redis-backed sliding window rate limits per endpoint category; Joi/Zod schema validation; DOMPurify sanitization.
FR-41	File Upload Security	ClamAV malware scan; MIME type verification; UUID-based naming; private S3 storage; EXIF stripping.
FR-42	Audit Logging	Immutable append-only log of all security-sensitive events with 2-year retention; exportable for compliance.
FR-43	Automated Due Date	Scheduled notifications at T-3, T-1, T=0 days; automatic Overdue

ID	Feature	Description
	Tracking	status assignment on deadline breach.
FR-44	Mandatory Evidence Collection	Owner and borrower must each upload min. 2 condition photos; digital in-app signatures required at handoff and return.
FR-45	Dispute Filing & Response	Owner dispute filing form with evidence upload; borrower 48-hour response window; auto-escalation on non-response.
FR-46	AI Photo Comparison & Chat Analysis	Vision model damage scoring (0-100) with annotated report; LLM chat log analysis for term verification and contradiction detection.
FR-47	Admin Arbitration & Resolution	Moderator/admin review with AI summary; binding resolution with deposit outcome and penalty assignment.
FR-48	Penalty & Compensation System	Tiered penalties (warning, suspension, ban) with trust score deductions; compensation obligations tracked with payment enforcement.
FR-49	Dispute Appeal Process	7-day appeal window with new evidence; reviewed by senior admin; outcome is final; full archive maintained for 3 years.

3. Non-Functional Requirements

ID	Requirement	Priority	Description
NFR-01	Scalability	High	System must handle 100K+ concurrent users and millions of item listings with horizontal scaling.
NFR-02	Performance	High	Search and recommendation engine must return results in under 500ms (95th percentile).
NFR-03	Security	Critical	Wallet/deposit handling must be encrypted; all endpoints protected by JWT + RBAC.
NFR-04	Reliability	High	Platform must maintain 99% uptime SLA with fault-tolerant dispute and wallet services.
NFR-05	Usability	Medium	UI must be mobile-responsive, WCAG 2.1 accessible, and require no training to use.
NFR-06	Data Integrity	Critical	All transaction and dispute records must be immutable and auditable.
NFR-07	Maintainability	Medium	Codebase follows modular, clean architecture with CI/CD pipeline and >80% test coverage.

4. Borrow & Exchange Workflow

The following describes the end-to-end workflow for a complete exchange transaction on LocalLoop, incorporating all FAANG-level safety and accountability mechanisms:

Step	Stage	Description
Step 1	Item Listing	Owner adds item with description, images, condition, borrow duration options, and damage/late-return rules.
Step 2	Borrow Request	Borrower selects item, views required security deposit and terms, accepts digital agreement to proceed.
Step 3	Deposit Lock	System dynamically calculates and locks borrower's deposit based on item value and trust score.
Step 4	Approval & Exchange	Owner approves request; both parties meet at a recommended Safe Exchange Zone for physical handoff.
Step 5	Active Borrow Tracking	Countdown timer activates; system tracks Active / Overdue / Completed status with auto-reminders.
Step 6	Item Return	Borrower submits proof (images/video); system compares condition against original listing.
Step 7	Late / Damage Handling	If overdue: penalty deducted, owner notified. If damaged: deposit deducted proportionally.
Step 8	Dispute Resolution	Either party can raise a dispute; evidence submitted; AI reviews and suggests verdict; admin finalizes.
Step 9	Trust & Penalty Update	Borrower or owner rating updated; penalty points deducted from responsible party's trust score.
Step 10	Deposit Release	Full or partial deposit refunded to borrower's wallet; transaction history updated for both parties.
Step 11	Status Transparency	All status updates (Borrowed / Returning / Under Dispute / Resolved) visible to both parties in real time.
Step 12	Dashboard & Analytics	Exchange recorded in user dashboard and admin analytics with fraud detection metrics updated.

4.1 Logical Flow Diagram

Item Listing → Borrow Request → Deposit Lock → Exchange (Safe Zone) → Return → On Time / Late / Damaged → Dispute (if any) → Resolution → Deposit Refund / Penalty → Dashboard & Analytics

5. System Entities & Data Models

The following core entities form the data model of LocalLoop:

Entity	Key Attributes
User	id, name, email, phone, trustScore, penaltyPoints, walletBalance, rating, activeExchanges, exchangeHistory
Item	id, ownerId, title, description, condition, images, borrowDurationOptions, rules, status, createdAt
BorrowRequest	id, itemId, borrowerId, requestedDuration, status, createdAt, approvedAt
Agreement	id, borrowRequestId, terms, depositAmount, acceptedAt, signature
Dispute	id, borrowRequestId, raisedBy, evidence, aiVerdict, adminVerdict, status, resolvedAt
Wallet / Deposit	id, userId, balance, lockedAmount, transactionHistory[]
Notification	id, userId, type, message, read, createdAt
SafeExchangeZone	id, name, location (lat/lng), address, verified
ChatSession	id, userId, language, messages[], intentHistory[], createdAt, lastActive
ChatMessage	id, sessionId, role (user/bot), content, intent, language, confidence, timestamp

6. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS / Chakra UI	Responsive SPA with component-based architecture
Frontend	Socket.io Client / Firebase SDK	Real-time notifications and live status updates
Backend	Node.js + Express.js	RESTful API server; handles all business logic
Backend	GraphQL (optional)	Flexible querying for complex nested data (items, exchanges)
Database	MySQL (Relational)	Relational data storage for items, users, exchanges, and disputes with structured schema and foreign key integrity
Cache	Redis	Session caching, deposit status, frequent query results
AI / ML	OpenAI GPT-4o API	AI Negotiation Assistant, dispute verdict suggestions, multilingual chatbot
AI / ML	langdetect / LLM language detection	Auto-detect user language for chatbot routing
AI / ML	Image Recognition API	Item condition comparison (before/after return)
Maps	Google Maps Platform API	Safe exchange zone discovery, location-based search
Auth	JWT + bcrypt	Stateless authentication and password hashing
Auth	Role-Based Access Control (RBAC)	Admin, Owner, Borrower permission layers
Notifications	Firebase Cloud Messaging / Socket.io	Push notifications and real-time WebSocket events
DevOps	Docker + CI/CD (GitHub Actions)	Containerised deployment and automated testing pipeline
Hosting	AWS / GCP / Railway	Cloud deployment with horizontal scaling support

7. System Architecture

7.1 Architecture Components

ID	Component	Technology	Responsibility
C-01	Client / Frontend	React SPA	Handles all user interactions: item browsing, borrow requests, dashboard, real-time status
C-02	API Gateway	Express.js	Routes requests to appropriate services; handles auth middleware and rate limiting
C-03	Auth Service	JWT + RBAC	Issues tokens, validates sessions, enforces role-based permissions
C-04	Item Service	MySQL	CRUD for items and skills; search/filter endpoints
C-05	Exchange Service	Node.js	Manages borrow requests, agreements, status transitions, and return tracking
C-06	Wallet / Deposit Service	MySQL + Redis	Locks/releases deposits, maintains transaction history, prevents double-spend
C-07	AI Service	OpenAI API	Runs negotiation prompts, suggests trade alternatives, analyzes dispute evidence
C-08	Notification Service	Firebase / Socket.io	Publishes real-time events to connected clients and mobile push notifications
C-09	Maps Service	Google Maps API	Returns nearby safe exchange zones based on both parties' locations
C-10	Dispute Engine	Node.js + AI Service	Orchestrates evidence collection, AI review, and admin finalization
C-11	Analytics Service	MySQL Aggregations (GROUP BY / Window Functions)	Powers user and admin dashboards with exchange stats, fraud signals, leaderboards

7.2 High-Level Architecture Layers

The system follows a layered microservice-inspired architecture:

- **Presentation Layer:** React SPA communicates with backend via REST/GraphQL and Socket.io for real-time updates.
- **API Layer:** Express.js API Gateway routes to domain services (Auth, Item, Exchange, Wallet, AI, Maps, Dispute, Analytics).
- **Data Layer:** MySQL as the primary relational store with Redis caching for hot-path queries and deposit state.
- **AI Layer:** OpenAI API calls are proxied through a dedicated AI Service module to handle prompt engineering and response parsing.
- **External Services:** Google Maps API, Firebase Cloud Messaging, and optional image recognition APIs.

7.3 Security Architecture

- All API endpoints protected by JWT Bearer token authentication.
- RBAC enforced at middleware level: Admin, Owner, and Borrower have distinct permission sets.
- Wallet service runs in an isolated context; deposit mutations require idempotency keys.

- All dispute evidence and transaction records are immutable once committed.
- HTTPS enforced; sensitive fields (passwords, tokens) hashed with bcrypt / never stored in plaintext.

7.4 Scalability Design

- MySQL read replicas and table partitioning for item and user data; dedicated replica for analytics queries.
- Redis pub/sub for horizontal scaling of Socket.io notification service.
- Stateless API servers allow auto-scaling behind a load balancer.
- AI service calls batched and cached where applicable to control OpenAI API costs.

8. AI & Smart Features

8.1 AI Negotiation Assistant

When a borrower and owner cannot agree on exchange terms, the AI Negotiation Assistant (powered by an LLM) steps in to:

- Analyze both parties' requirements and item values
- Suggest fair deposit amounts based on item condition and market value
- Propose alternative exchange durations or trade-for-trade options
- Draft counter-offer messages in natural language

8.2 AI-Assisted Dispute Resolution

When a dispute is raised, the AI Dispute Engine:

- Receives both parties' submitted evidence (images, descriptions, timestamps)
- Compares before/after item condition using image analysis
- Reviews borrow agreement terms against claimed violations
- Generates a structured verdict recommendation with confidence score
- Admin reviews AI recommendation before finalizing; full audit trail maintained

8.3 Multi-User Exchange Chain Detection

A graph-based algorithm detects circular exchange opportunities across users:

- User A has Item X (wanted by B), User B has Item Y (wanted by C), User C has Item Z (wanted by A)
- System notifies all parties of the potential 3-way exchange chain
- Single coordinated exchange reduces friction and deposit complexity

9. Core API Endpoints

Representative REST API endpoints for the LocalLoop platform:

Method	Endpoint	Access	Description
POST	/api/auth/register	User	Register new user account
POST	/api/auth/login	User	Authenticate and receive JWT
GET	/api/items	User	Search and filter item listings
POST	/api/items	Owner	Create new item listing
POST	/api/borrow/request	Borrower	Submit a borrow request
PUT	/api/borrow/:id/approve	Owner	Approve a borrow request
POST	/api/borrow/:id/return	Borrower	Submit return proof
POST	/api/disputes	User	Raise a dispute with evidence
GET	/api/wallet	User	View wallet balance and history
GET	/api/zones/nearby	User	Get safe exchange zones near location
GET	/api/admin/analytics	Admin	View platform-wide analytics
GET	/api/admin/disputes	Admin	View and resolve pending disputes

10. Multilingual AI Chatbot

LocalLoop integrates a fully multilingual AI chatbot powered by a Large Language Model (LLM) to assist users throughout their exchange journey — from discovering items to resolving disputes — in their native language.

10.1 Feature Overview

- **Always-on chat widget accessible from every screen of the application**
- **Supports 17 languages including all major Indian regional languages**
- **Auto-detects user's language from browser locale, profile preference, or first message**
- **Maintains conversation context across multiple turns within a session**
- **Seamlessly hands off to a human admin agent for complex dispute queries**
- **Responds in under 2 seconds for common intents using cached prompt templates**

10.2 Supported Languages

The chatbot supports the following 17 languages at launch:

Language	ISO Code	Coverage Note
Hindi	hi	India — primary user base
English	en	Default / international
Tamil	ta	South India
Telugu	te	South India
Bengali	bn	East India / Bangladesh
Marathi	mr	Maharashtra
Gujarati	gu	Gujarat
Punjabi	pa	Punjab region
Kannada	kn	Karnataka
Malayalam	ml	Kerala
Urdu	ur	Pakistan / North India
Spanish	es	Latin America / Spain
French	fr	France / Africa
Arabic	ar	Middle East / North Africa
Portuguese	pt	Brazil / Portugal
Indonesian	id	Southeast Asia
Chinese (Simplified)	zh	China / Singapore

10.3 Supported Intents

The chatbot is trained to handle the following primary intents:

Intent	Example Trigger	Bot Action
how_to_borrow	How do I borrow an item?	Explains the 12-step borrow workflow in the user's language
check_deposit	What is my deposit amount?	Fetches locked deposit from Wallet Service and explains calculation
exchange_status	What is the status of my exchange?	Queries Exchange Service and returns real-time borrow status
raise_dispute	How do I raise a dispute?	Guides user through dispute submission flow with step-by-step instructions
find_safe_zone	Where is the nearest safe exchange zone?	Calls Google Maps API and returns closest verified exchange locations
trust_score	Why did my trust score change?	Explains recent rating/penalty events that affected the trust score
return_item	How do I return an item?	Explains proof submission (image/video) and return confirmation steps
wallet_balance	What is my wallet balance?	Returns current wallet balance and locked deposit summary
platform_faq	General platform questions	Covers fees, rules, policies, account management in user's language
ai_negotiation	Help me negotiate this trade	Triggers AI Negotiation Assistant with context from active request
language_switch	Change language	Switches chatbot response language without losing conversation context

10.4 Technical Architecture

10.4.1 Language Detection & Routing

- Browser locale header and user profile language preference are checked first.
- If no preference is set, the first user message is passed to a language detection model (e.g., langdetect / OpenAI language identification prompt).
- Detected language is stored in the chat session and used for all subsequent LLM system prompts.
- Users can switch language at any time via the /language command or settings toggle.

10.4.2 LLM Integration

- Primary model: OpenAI GPT-4o (or GPT-4 Turbo) via the Anthropic API proxy.
- System prompt is dynamically constructed with: (1) platform context, (2) user's active exchange data, (3) target language instruction.
- Example system prompt injection: "You are the LocalLoop assistant. Respond only in [LANGUAGE]. The user's current borrow status is [STATUS]. Their wallet balance is [BALANCE]."
- Token-efficient templates cached in Redis for high-frequency intents (FAQ, how-to-borrow, wallet balance).

10.4.3 Live Data Integration

The chatbot backend is connected to all core LocalLoop services to provide real-time, personalised responses:

Service	Data Fetched	Used For
Exchange Service	Borrow status, due date, return proof status	Status queries, reminders
Wallet Service	Balance, locked deposit, recent transactions	Deposit & balance queries
Google Maps API	Nearby safe exchange zones	Location-based help
Dispute Engine	Open disputes, evidence submitted, verdict status	Dispute guidance
User Profile Service	Trust score, penalty history, ratings	Trust score explanations
Item Service	Listed items, condition, owner details	Browse & recommendation help

10.4.4 Human Handoff

- If the chatbot confidence score falls below a threshold (e.g., <0.75), it offers to escalate to a human admin.
- Conversation transcript is passed to the admin dispute dashboard with full context.
- Handoff notification sent to admin via the Notification Service in real time.

10.5 Non-Functional Requirements

Requirement	Target
Response Latency	< 2 seconds for cached intents; < 5 seconds for LLM-generated responses
Language Accuracy	> 95% intent detection accuracy across all 17 supported languages
Uptime	99.9% — chatbot service deployed independently from core API
Context Window	Maintain last 10 conversation turns per session for context continuity
Data Privacy	No conversation history stored beyond session unless user opts in
Fallback Behaviour	If LLM API is unavailable, serve cached FAQ responses in detected language

10.6 UI/UX Design

- Floating chat bubble in bottom-right corner on all pages.
- Language flag/selector visible at the top of the chat window with dropdown for manual override.
- Typing indicator shown while LLM is generating a response.
- Quick-reply chips for common actions: 'Check my deposit', 'Find safe zone', 'Raise a dispute', 'Return item'.
- Chat history persisted locally (sessionStorage) so users can scroll back within the same session.
- Dark mode support matching the main application theme.

10.7 Chatbot API Endpoints

Method	Endpoint	Access	Description
POST	/api/chatbot/message	User	Send message; returns AI response in detected/selected language

Method	Endpoint	Access	Description
POST	/api/chatbot/language	User	Explicitly set preferred language for chatbot session
GET	/api/chatbot/history	User	Retrieve conversation history for active session
POST	/api/chatbot/feedback	User	Submit thumbs up/down on a chatbot response for fine-tuning
GET	/api/admin/chatbot/analytics	Admin	View language usage stats, unanswered queries, drop-off rates

12. User-Friendly Feature Set

LocalLoop is enriched with 20 user-centric features that make the platform intuitive, inclusive, and engaging for all community members regardless of language, ability, or device. These features extend the core exchange workflow with communication, personalisation, safety, and collaboration tools.

12.1 Feature Register

All 20 features with ID, priority, and full description:

ID	Feature	Priority	Description
UF-01	Real-Time Language Translation	High	Instantly translate messages between users. AI auto-detects the source language and renders the translated message inline without the recipient needing to change any settings.
UF-02	Voice Input & Output	High	Users can speak instead of typing using the Web Speech API / Whisper. AI responds with synthesised voice (TTS) in the user's preferred language.
UF-03	Personalized Recommendations	High	AI analyses user activity, browsing history, and interests to surface relevant items, posts, local events, and services using a collaborative filtering engine.
UF-04	Chat Summarization	Medium	LLM summarises long conversations or articles into 3–5 bullet points so users can quickly catch up without reading the full thread.
UF-05	Context-Aware Responses	High	AI retains user preferences and past interactions across sessions (with consent) to personalise every chatbot and assistant response.
UF-06	Emoji, GIF & Sticker Support	Medium	Full Unicode emoji picker, GIPHY-powered GIF search, and custom LocalLoop sticker packs to make chatting expressive and fun.
UF-07	Image & Media Recognition	High	Users upload images; AI identifies objects, landmarks, item conditions, or extracts and translates text from images (OCR + Vision API).
UF-08	Quick Reply Suggestions	High	AI generates 3–5 contextual reply chips below each incoming message, plus action shortcuts (e.g., 'Approve Request', 'View Deposit').
UF-09	Community Threads & Polls	Medium	Users can start topic discussions and run polls; AI summarises thread responses and highlights consensus or majority opinion.
UF-10	Sentiment & Mood Detection	Medium	AI analyses message tone in real time, detects frustration or distress, and adapts its reply style to be empathetic and de-escalating.
UF-11	Gamification & Rewards	Medium	Badges, XP points, leaderboards, and weekly challenges (e.g., 'Complete 5 exchanges') to boost engagement and user retention.
UF-12	Accessibility Options	High	Dark/light mode toggle, adjustable font sizes, high-contrast theme, keyboard navigation, screen-reader support, and full voice command control.
UF-13	Safety & Moderation	Critical	AI scans messages and images for hate speech, spam, and inappropriate content using a moderation model. Users can report or block others; flagged content queued for admin review.
UF-14	Offline Chat Mode	Low	Service Worker caches recent conversations and FAQ

ID	Feature	Priority	Description
			responses. Users can read history and compose messages offline; synced when connectivity resumes.
UF-15	Multi-Platform Sync	High	Conversations, preferences, and state are synced in real time across web, Android, and iOS via Firebase Firestore, so users resume exactly where they left off.
UF-16	Smart Notifications	High	AI personalises notification frequency and content per user — suppressing low-priority alerts, batching updates, and surfacing urgent events (overdue item, new dispute) instantly.
UF-17	AI-Powered FAQs & Help	High	Semantic search over a curated knowledge base gives instant, accurate answers to platform questions without opening a support ticket.
UF-18	Topic-Based Chatrooms	Medium	Users create or join rooms by interest or location. AI suggests active rooms based on profile, and summarises room conversations for new joiners.
UF-19	Content Moderation with Auto-Tagging	High	AI automatically tags posts, images, and messages with categories for search and discovery, and proactively flags policy-violating content before it is visible to others.
UF-20	Collaborative Tools	Medium	AI assists in creating shared notes, to-do lists, and lightweight documents within community groups, with real-time co-editing and version history.

12.2 Technical Implementation Specifications

Recommended technology and implementation approach for each feature group:

Feature Group	Technology	Implementation Notes
Translation (UF-01)	Google Translate API / DeepL API	Language detection via langdetect; translate on-the-fly per message; store original + translated pair
Voice I/O (UF-02)	Web Speech API + OpenAI Whisper (STT) + Google TTS / ElevenLabs	Whisper transcribes audio to text; LLM processes; TTS renders audio response
Recommendations (UF-03)	Collaborative filtering + OpenAI Embeddings	User vectors stored in Redis; cosine similarity matching against item/event embeddings
Summarization (UF-04)	OpenAI GPT-4o	Prompt: summarise thread in ≤5 bullets in [LANGUAGE]; token-limited to last 50 messages
Context Memory (UF-05)	Redis (session) + MySQL (long-term, consent-gated)	Session context injected into every LLM system prompt; persistent profile stored with user opt-in
Emoji / GIF (UF-06)	GIPHY API + Unicode emoji standard	Frontend emoji picker; GIF search via GIPHY; custom sticker assets served from CDN
Image Recognition (UF-07)	OpenAI Vision API / Google Cloud Vision	Multipart upload → Vision API → structured JSON response → rendered as chat card
Quick Replies (UF-08)	OpenAI GPT-4o function calling	Last 3 messages passed to suggestion prompt; returns JSON array of chips; rendered as buttons
Threads & Polls (UF-09)	MySQL + LLM summarisation	Poll results stored in MySQL; AI summarises thread on demand or after 24 hours
Sentiment (UF-10)	OpenAI GPT-4o / HuggingFace sentiment model	Per-message sentiment score; tone flag stored on message doc; empathy mode activates at score < -0.5
Gamification (UF-11)	MySQL + Redis leaderboard sorted sets	XP awarded on exchange completion, on-time return, positive rating; badges stored on user profile
Accessibility (UF-12)	CSS custom properties + ARIA + Web Speech API	Theme stored in localStorage; font scale via CSS variables; voice commands routed to chatbot service
Moderation (UF-13)	OpenAI Moderation API + custom classifier	Every message/image passed through moderation endpoint before delivery; score threshold triggers hold queue
Offline Mode (UF-14)	Service Worker + IndexedDB	Workbox caches last 100 messages and FAQ JSON; outbox queues offline-composed messages for sync
Multi-Platform Sync (UF-15)	Firebase Firestore real-time sync	Firestore document per conversation; onSnapshot listener updates all open sessions within 200ms
Smart Notifications (UF-16)	Firebase Cloud Messaging + ML preference model	User notification preference vector updated daily; urgency classifier routes critical alerts immediately
AI FAQs (UF-17)	OpenAI Embeddings + vector search (Pinecone / pgvector)	FAQ articles embedded at deploy time; user query embedded at runtime; top-3 nearest articles returned
Chatrooms (UF-18)	MySQL + Socket.io rooms + LLM	Room topics stored as tags; AI room-join suggestion based on user interest embedding similarity
Auto-Tagging (UF-19)	OpenAI GPT-4o + Vision API	Tag extraction prompt run on every post/image; tags stored as indexed array on content document
Collaborative Tools (UF-20)	Operational Transform (OT) / Yjs CRDT + MySQL	Shared document state managed via Yjs; AI assist triggered by /ai command in doc editor

12.3 Feature Group Architecture

12.3.1 Communication Layer

Features UF-01 (Translation), UF-02 (Voice), UF-06 (Emoji/GIF), UF-08 (Quick Replies) all operate at the message delivery layer, sitting between the Socket.io real-time transport and the chat UI renderer. Each message passes through a pipeline:

- Inbound message received via Socket.io
- Language detection → translate if source != recipient language (UF-01)
- Sentiment analysis → adjust response tone flags (UF-10)
- Moderation check → hold or deliver (UF-13)
- Auto-tag extraction → store tags on message document (UF-19)
- Quick reply suggestions generated asynchronously → pushed to client (UF-08)
- Message delivered to recipient with translated text, sentiment tag, and reply chips

12.3.2 Personalisation Engine

Features UF-03 (Recommendations), UF-05 (Context Memory), UF-16 (Smart Notifications), and UF-17 (AI FAQs) share a common Personalisation Engine:

- User activity events (item views, exchanges, searches) streamed to a Redis event queue.
- Batch processor runs every 6 hours to update user interest vectors using OpenAI Embeddings.
- At query time, user vector is compared against item/event/FAQ embeddings (cosine similarity).
- Top-N results injected into chatbot context, notification payloads, and recommendation cards.

12.3.3 Safety & Moderation Pipeline

Features UF-13 (Safety & Moderation) and UF-19 (Auto-Tagging) form a unified content pipeline:

- Every text message and image upload is routed through the OpenAI Moderation API before delivery.
- Score thresholds: SAFE (< 0.5) → deliver; WARN (0.5–0.8) → add warning label; BLOCK (> 0.8) → hold for admin.
- Auto-tagging LLM prompt runs in parallel (non-blocking) and attaches tags to the content document.
- Blocked content placed in admin moderation queue with full context for review within 24 hours.

12.3.4 Gamification & Engagement Engine

Feature UF-11 (Gamification) is powered by an event-driven reward engine:

- Exchange events (request approved, on-time return, positive rating received) emit reward events.
- Reward engine consumes events, calculates XP delta, and writes idempotently to user profile.
- Badge criteria evaluated on every XP update; newly earned badges trigger a push notification.
- Leaderboard maintained as a Redis sorted set, refreshed in real time.

12.4 Non-Functional Requirements

Requirement	Target
Translation Latency	< 300ms per message for real-time translation
Voice Transcription Accuracy	> 95% word accuracy for supported languages
Recommendation Freshness	Recommendation vectors refreshed every 6 hours
Moderation Latency	< 150ms — moderation check completes before message delivery
Offline Cache Size	Last 100 messages + FAQ JSON; max 50MB IndexedDB quota

Requirement	Target
Sync Latency	Multi-platform sync within 200ms on active connection
Gamification Consistency	XP and badge awards idempotent; deduplication via event IDs
Accessibility Compliance	WCAG 2.1 Level AA across all UI components

12.5 Updated Tech Stack Additions

Layer	Technology Added	Purpose
Translation	Google Translate API / DeepL API	Real-time message translation (UF-01)
Voice	OpenAI Whisper (STT) + Google TTS / ElevenLabs	Voice input transcription and TTS output (UF-02)
Vision	OpenAI Vision API / Google Cloud Vision	Image recognition and OCR (UF-07)
Vector Search	Pinecone / pgvector (PostgreSQL)	Semantic FAQ search and recommendation engine (UF-03, UF-17)
Offline	Service Worker + IndexedDB (Workbox)	Offline chat mode and background sync (UF-14)
Sync	Firebase Firestore	Real-time multi-platform conversation sync (UF-15)
Media	GIPHY API + CDN (Cloudflare)	GIF search and sticker asset delivery (UF-06)
Collab	Yjs CRDT library	Real-time collaborative document editing (UF-20)
Moderation	OpenAI Moderation API	Content safety pipeline for messages and images (UF-13, UF-19)

13. Security Architecture

Security is a first-class concern for LocalLoop, which handles sensitive user data, financial deposits, and community trust. The model follows defence-in-depth, covering authentication, data protection, network security, file handling, and compliance.

13.1 Security Requirements Register

ID	Requirement	Priority	Description
SEC-01	JWT / OAuth 2.0 Authentication	Critical	All sessions authenticated via signed JWT tokens (RS256) or OAuth 2.0 provider flows; access tokens expire in 15 minutes with refresh rotation.
SEC-02	Multi-Factor Authentication (MFA)	High	Email OTP + SMS as second factor; TOTP (Google Authenticator) as optional third; enforced for all admin accounts.
SEC-03	Role-Based Access Control (RBAC)	Critical	Three roles: User, Moderator, Admin. Distinct permission sets enforced at middleware level on every API route.
SEC-04	AES-256 / RSA Data Encryption	Critical	PII fields encrypted at rest with AES-256-GCM; per-document keys managed via AWS KMS; RSA-2048 for key exchange.
SEC-05	Password Hashing (Argon2 / bcrypt)	Critical	Argon2id with OWASP-recommended params; bcrypt (cost 12) as fallback. Plaintext passwords never stored or logged.
SEC-06	Parameterized Queries / ORM	Critical	All DB interactions via Sequelize ORM or parameterized prepared statements; raw query construction prohibited to prevent SQL injection.
SEC-07	HTTPS / TLS 1.3 Enforcement	Critical	All traffic over TLS 1.3; HTTP redirected to HTTPS; HSTS header with 1-year max-age enforced.
SEC-08	Rate Limiting & Throttling	High	100 req/min per authenticated user; 20 req/min unauthenticated. Redis-backed sliding window counters.
SEC-09	Input Validation & Sanitization	Critical	All inputs validated with Joi/Zod schemas; HTML sanitized via DOMPurify; SQL injection blocked via parameterized queries and ORM-enforced schema validation.
SEC-10	Audit Logging	High	Every user action logged with timestamp, IP, and user ID to an immutable append-only audit trail retained for 2 years.
SEC-11	File Upload Security	High	Files scanned with ClamAV; MIME type verified server-side; max 10MB; stored in private S3 bucket outside webroot.
SEC-12	Security Headers (Helmet.js)	High	CSP, X-Frame-Options, X-Content-Type-Options, HSTS, Referrer-Policy, and Permissions-Policy applied on every response.
SEC-13	CSRF Protection	High	CSRF tokens required for all state-changing requests; SameSite=Strict cookie attribute enforced.
SEC-14	Dependency Scanning	Medium	Automated npm audit and Snyk scans on every CI/CD run; critical CVEs block deployment.
SEC-15	Penetration Testing	Medium	Quarterly external pen tests; OWASP Top 10 checklist reviewed before every major release.

13.2 Authentication & Authorisation

13.2.1 JWT Token Lifecycle

- Access Token: short-lived (15 min), signed with RS256, contains userId, role, and sessionId.
- Refresh Token: long-lived (7 days), stored as HttpOnly SameSite=Strict cookie; rotated on every use.
- Token revocation: Redis blocklist stores invalidated tokens until natural expiry.
- OAuth 2.0 social login supported (Google, Apple) as alternative to email/password.

13.2.2 Multi-Factor Authentication Flow

- Step 1: User submits email + password — server returns a partial session token (MFA pending).
- Step 2: OTP sent via email (primary) or SMS (secondary); valid for 10 minutes, single-use.
- Step 3: User submits OTP — full JWT access token issued on success.
- Step 4: MFA bypass codes generated at enrollment; stored hashed in DB for account recovery.
- Admin accounts: MFA is mandatory and cannot be disabled.

13.2.3 Role-Based Access Control Matrix

Permission	User	Moderator	Admin
View item listings	Yes	Yes	Yes
Submit borrow request	Yes	Yes	Yes
List / edit own items	Yes	Yes	Yes
Rate & review users	Yes	Yes	Yes
Raise a dispute	Yes	Yes	Yes
Moderate reported content	No	Yes	Yes
View all disputes	No	Yes	Yes
Issue warnings / temp bans	No	Yes	Yes
Access admin analytics	No	No	Yes
Manage user accounts	No	No	Yes
Configure platform settings	No	No	Yes
View full audit logs	No	No	Yes

13.3 Data Protection

13.3.1 Encryption at Rest

- MySQL column-level encryption on: phone number, national ID, bank account reference, residential address.
- AES-256-GCM with per-row keys; master key stored in AWS KMS.
- Wallet transaction amounts stored in plaintext for analytics; PII fields within transactions encrypted.

13.3.2 Encryption in Transit

- TLS 1.3 enforced for all client-server and service-to-service communication.
- Internal microservice calls over mTLS within the Kubernetes cluster.

- Database connections require TLS certificates; unencrypted connections rejected by MySQL server config (require_secure_transport=ON).

13.3.3 Password Security

- Argon2id: memory=65536KB, iterations=3, parallelism=4 (OWASP recommended).
- bcrypt cost factor 12 used as fallback in environments where Argon2 is unavailable.
- Minimum 8 characters; at least one uppercase, one digit, one symbol enforced.
- Compromised password check via HavelBeenPwned k-Anonymity API on registration and change.

13.4 Network & API Security

13.4.1 Rate Limiting Strategy

Endpoint Category	Limit	Window	Action on Breach
Authentication (login, OTP)	5 requests	15 minutes	429 + 15-min lockout
General API (authenticated)	100 requests	1 minute	429 with Retry-After header
General API (unauthenticated)	20 requests	1 minute	429 + CAPTCHA challenge
File Upload	10 uploads	1 hour	429 + admin alert if repeated
Dispute Filing	3 disputes	24 hours	Blocked + moderator notification
Admin Endpoints	500 requests	1 minute	Alert logged; access reviewed

13.4.2 Input Validation & Injection Prevention

- All request bodies validated against strict Joi/Zod schemas; unexpected fields stripped.
- SQL keywords and operators sanitized; parameterized queries used exclusively; ORM prevents dynamic query construction.
- HTML content sanitized with DOMPurify before storage; raw HTML never rendered from user input.
- File path inputs canonicalized and checked against allowed directories to prevent path traversal.

13.4.3 Security Headers

- Content-Security-Policy: default-src 'self'; trusted CDNs only for scripts.
- X-Frame-Options: DENY — prevents clickjacking attacks.
- X-Content-Type-Options: nosniff — prevents MIME type sniffing.
- Strict-Transport-Security: max-age=31536000; includeSubDomains; preload.
- Permissions-Policy: geolocation, microphone, camera disabled by default.

13.5 File Upload Security

Control	Implementation	Notes
Allowed File Types	JPEG, PNG, WebP, MP4 (video proof only)	MIME type verified server-side
Maximum File Size	Images: 10MB Videos: 50MB	Enforced at API gateway before storage write
Malware Scanning	ClamAV real-time scan on upload	Infected files rejected with 422; uploader notified

Control	Implementation	Notes
Storage Location	Private AWS S3 bucket (not public)	Pre-signed URLs with 15-min expiry for access
File Naming	UUID-based filenames only	Original filenames stripped to prevent path traversal
Image Metadata	EXIF data stripped on ingest	Prevents location data leakage from user photos

13.6 Audit Logging

- Events logged: login, logout, failed login, MFA, password change, item CRUD, borrow lifecycle, dispute actions, admin actions, wallet mutations.
- Log schema: { eventType, userId, ipAddress, userAgent, timestamp, resourceId, outcome }.
- Logs written to an append-only MySQL audit_log table with 2-year retention enforced via scheduled archival jobs.
- Critical events trigger real-time alerts to the security monitoring dashboard.
- Logs exportable by admin in JSON/CSV for compliance and forensic review.

14. Enhanced Dispute Handling System

LocalLoop's dispute system is a multi-stage, evidence-driven workflow that protects both owners and borrowers through automated tracking, mandatory photo evidence, AI-assisted analysis, and human arbitration — ensuring fair and transparent outcomes every time.

14.1 Dispute Requirements Register

ID	Requirement	Description
DS-01	Automated Due Date Tracking	System sends reminders at T-3 days, T-1 day, and T=0. Auto-marks item Overdue if not returned by deadline.
DS-02	Late Return Alerts	Simultaneous notifications to borrower and owner at first missed deadline; escalating alerts every 24 hours.
DS-03	Mandatory Photo Evidence (Borrow)	Owner uploads min. 2 photos before exchange is confirmed; photos timestamped and stored immutably in S3.
DS-04	Mandatory Photo Evidence (Return)	Borrower uploads min. 2 return photos; required before exchange can be marked Complete.
DS-05	Digital Signatures	Both parties sign in-app (canvas-based) at handoff and return; stored with the Agreement record.
DS-06	Dispute Filing by Owner	Owner files dispute if item overdue, returned damaged, or not returned; form collects evidence and desired resolution.
DS-07	Borrower Response Window	Borrower has 48 hours to respond with counter-evidence; non-response treated as no contest.
DS-08	AI Photo Comparison	Vision model compares before/after photos; outputs damage score (0-100) and annotated difference report.
DS-09	AI Chat Log Analysis	LLM reviews in-platform messages to identify agreed terms, admissions, and contradictions.
DS-10	Automatic Escalation	If borrower silent after 48h, dispute auto-escalates to admin with AI recommendation pre-populated.
DS-11	Community Arbitration	Moderator/admin reviews all evidence and AI summary before issuing a binding resolution.
DS-12	Penalty System	Tiered penalties: warning, temp suspension, permanent ban. Trust score deducted based on severity.
DS-13	Compensation Tracking	Compensation obligations tracked on borrower profile; payment deadline enforced with privilege suspension on non-payment.
DS-14	Deposit Release Logic	Full/partial/zero deposit release based on outcome; requires admin confirmation or auto-trigger on resolution.
DS-15	Dispute Analytics	Admin dashboard: open disputes, resolution times, AI accuracy rates, false-claim stats, top offenders.

14.2 End-to-End Dispute Workflow

Step	Stage	Action	Responsible
D-01	Trigger	Owner files dispute OR system auto-triggers at T+48h overdue	Platform
D-02	Notification	Borrower notified via push, email, in-app; 48h response window begins	System
D-03	Evidence Submission	Both parties upload additional evidence (photos, messages, receipts)	Both Parties
D-04	AI Assessment	Vision model compares before/after; LLM analyses chat logs; score generated	AI Service
D-05	AI Summary	Structured verdict recommendation produced with confidence score and reasoning	AI Service
D-06	Admin Review	Moderator reviews AI summary and raw evidence; may request more info	Moderator/Admin
D-07	Resolution Issued	Binding decision: fault assigned, deposit outcome determined	Admin
D-08	Penalty Applied	Trust score updated, deposit released/forfeited, suspension triggered	System
D-09	Compensation Tracked	Payment obligation logged if compensation owed; deadline set	System
D-10	Appeal Window	Penalised party has 7 days to appeal with new evidence	Platform
D-11	Record Archived	Full dispute record archived immutably for 3 years	System

14.3 Automated Tracking & Alerts

14.3.1 Due Date Notification Schedule

Trigger	Recipient(s)	Channel	Message
T-3 days before due	Borrower	Push + In-App	Reminder: your borrowed item is due in 3 days. Please plan your return.
T-1 day before due	Borrower	Push + Email + In-App	Urgent: item due tomorrow. Upload return proof in the app.
T=0 (due date)	Borrower + Owner	Push + Email + SMS	Item return due today. Owner notified. Please return immediately.
T+1 day overdue	Borrower + Owner	Push + Email + SMS	Item now OVERDUE. Dispute filing now available to owner.
T+2 days overdue	Borrower + Owner + Admin	Push + Email + Alert	Second overdue notice. Auto-dispute escalation in 24 hours.
T+3 days overdue	Admin	Dashboard Alert	Auto-dispute triggered. AI assessment running. Admin review required.

14.3.2 Automatic Status Transitions

- Active => Overdue: borrow duration expires without return proof upload.
- Overdue => Disputed: T+3 days with no borrower response triggers auto-dispute.
- Disputed => Under Review: admin picks up dispute from moderation queue.
- Under Review => Resolved: admin issues final binding decision.
- Resolved => Archived: 7 days after resolution once appeal window closes.

14.4 Evidence Collection System

14.4.1 Mandatory Photo Requirements

- Before Exchange: owner uploads min. 2 photos from multiple angles; timestamped and stored as immutable originals.
- On Return: borrower uploads min. 2 matching-angle photos; system shows original angles as reference overlays.
- Video option: both parties may upload up to 60-second video (max 50MB) for high-value items.
- All photos displayed side-by-side in the dispute interface for direct visual comparison.

14.4.2 Digital Signatures

- Canvas-based in-app signature captured from both parties at exchange and return.
- Signatures stored as base64 images on the Agreement record with device fingerprint and timestamp.
- Both parties receive a PDF copy of the signed agreement via email immediately after exchange.

14.5 AI-Assisted Moderation

14.5.1 AI Photo Comparison

- Before/after photos passed to OpenAI Vision API with a structured damage assessment prompt.
- Output: damage score (0-100), affected areas identified, confidence level.
- Score bands: 0-10 (No damage), 11-30 (Minor), 31-60 (Moderate), 61-100 (Major/Destroyed).
- AI report displayed to admin with annotated images and score breakdown.

14.5.2 AI Chat Log Analysis

- LLM reviews last 200 messages of the in-platform conversation between parties.
- Identifies: agreed return dates, explicit condition acknowledgements, damage admissions, contradictions.
- Produces structured summary: agreed terms, key quotes, sentiment timeline, contradiction flags.
- Supporting evidence only; never the sole basis for a penalty decision.

14.5.3 AI Verdict Recommendation

The AI produces a structured recommendation presented to the admin before final review:

Field	Example Output
Damage Assessment	Score: 45/100 — Moderate damage detected on upper surface
Photo Comparison	3 areas of difference identified vs. original condition photos
Chat Evidence	Borrower acknowledged item was dropped on Day 4 of borrow period
Agreed Terms	Return date and deposit amount confirmed in chat transcript
Recommended Outcome	Partial deposit deduction: 40% — borrower at fault for moderate damage
Confidence Score	78% — moderate confidence; recommend admin review of flagged area

14.6 Penalty & Compensation System

14.6.1 Penalty Matrix

Offence	Platform Action	Trust Score Impact	Deposit Impact
First Late Return (1-3 days)	Warning notification	-5 trust points	None
Second Late Return	3-day borrowing suspension	-10 trust points	None
Chronic Late Return (3+)	30-day borrowing suspension	-25 trust points	None
Minor Damage (under 20%)	Warning + partial deduction	-10 trust points	Proportional deduction
Major Damage (20-80%)	7-day suspension	-30 trust points	Proportional deduction
Item Destroyed / Missing	Permanent ban (appealable)	-100 trust points	Full deposit forfeit
False Damage Claim (owner)	Warning issued to owner	-15 trust points	Claim dismissed
Repeat False Claims (owner)	Owner listing privileges suspended	-40 trust points	None

14.6.2 Compensation Tracking

- When admin assigns compensation, a compensation obligation is created on the borrower's profile.
- Borrower notified with payment deadline (default: 7 days from resolution).
- In-platform wallet payment: automatically marked Settled on receipt.
- Unpaid after deadline: collections flag added; listing and borrowing privileges suspended.
- Full history visible to admin; borrower sees own obligations on their dashboard.

14.7 Appeal Process

- Penalised party has a 7-day window after resolution to file an appeal with new evidence.
- Senior admin (separate from original reviewer) handles all appeals.
- Appeal outcome is final; no further appeals on the same dispute.
- If upheld: penalty reversed, deposit corrected, opposing party notified.
- If rejected: original decision stands; appeal record appended to dispute archive.

15. Project Repository & Deliverables

15.1 GitHub Repository

Repository: github.com/RajputSivam/LocalLoop-Community-Exchange-Platform

Branch: main

Structure:

- /frontend — React.js application
- /backend — Node.js + Express.js API server
- LocalLoop_Features.pdf — Original feature specification
- NeighborShare_User_Guide_v2.pdf — User guide documentation

15.2 Deliverables Checklist

Deliverable	Status	Notes
Multilingual AI Chatbot Specification	Complete	Section 10
User-Friendly Features Specification (20 Features)	Complete	Section 12
Security Architecture & Requirements	Complete	Section 13
Enhanced Dispute Handling System	Complete	Section 14
Software Requirements Specification (SRS)	Complete	This document
Technology Framework / Stack Table	Complete	Section 6
System Architecture Document	Complete	Section 7
FAANG-Level Workflow (12-Step)	Complete	Section 4
AI Features Specification	Complete	Section 8
Core API Endpoint Reference	Complete	Section 9
Data Model / Entity Reference	Complete	Section 5
Frontend (React.js)	In Progress	GitHub: /frontend
Backend (Node.js + Express)	In Progress	GitHub: /backend

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